

All Talked Out:
Retail Losses and Loss-Salience Saturation in
Online Investor Discussion

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Abstract

Retail losses often trigger negative online discussion, but losses may stop generating much new negativity once discussion is already saturated. This paper studies that margin using KRX individual-investor flows, stock prices, and a daily panel of 99 Korean retail message boards. We construct rolling 20-day retail long-side unrealized loss states from observed individual-investor lots. Retail loss states are associated with higher next-day negative discussion in lower-Neuroticism states, but this response attenuates when lagged board-day Neuroticism is already high. The attenuation remains under anchored cumulative and daily-deterioration alternatives. We interpret this pattern as reduced-form evidence of loss-salience saturation. Low-Neuroticism states are less saturated states in which retail losses remain capable of generating negative discussion, whereas high-Neuroticism states are more saturated states in which the marginal response to further losses has flattened. Consistent with this interpretation, high-Neuroticism days are contemporaneously negative but exhibit weaker subsequent diffusion, and current negative discussion has a stronger cumulative-return association in low-Neuroticism states than in high-Neuroticism states. We therefore interpret board-day Neuroticism as a language-based behavioral state variable for retail loss-salience saturation, not as a stand-alone scalar sentiment measure or a structural estimate of investor loss aversion.

Keywords: retail investor losses; loss-salience saturation; textual analysis; investor sentiment; behavioral state variable; sentiment diffusion; retail investor discussion

JEL codes: G41; G14; G12; D83; C55

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1 Introduction

Retail investors often process losses in public. Stock message boards therefore contain not only opinions about fundamentals, but also traces of how retail investors react when recent positions move against them. This makes online discussion a natural place to study how losses become socially and market relevant. Yet the level of negative talk is an incomplete object. A board can be negative because an adverse state is still unfolding, or because the same loss-related concerns have already been repeatedly expressed and absorbed. This distinction matters because market-relevant discussion depends on remaining response capacity, not only on current negativity. In the first case, retail loss states can still draw attention, generate new negative language, and carry information into subsequent discussion and prices. In the second case, the board may already be saturated: investors have already articulated the loss and attention has already concentrated, so a similar retail loss state may add little new language or market-relevant content. For market relevance, the relevant margin is therefore not only how negative discussion is, but whether it remains responsive to retail losses.

Most text-based measures are not designed to capture this margin. They summarize discussion with a scalar tone, sentiment, attention, or opinion index. Such measures are useful for measuring how negative a board is, but they do not indicate whether loss-related negativity still has marginal response capacity. The same current negative share can therefore represent either an early, diffusion-active state with remaining response capacity or a late, self-limiting state in which additional losses have little marginal effect.

The empirical object is the marginal response of negative discussion to retail loss states, conditional on the current level of negativity. We study that object using daily discussion from 99 Korean retail stock boards and KRX individual-investor flow data. The flow data let us move beyond simple price declines and measure whether recently observed retail long-side lots are in loss territory. We pair this retail loss-state measure with board-day Neuroticism, the share of messages written in Neuroticism-like language on a given stock board and trading day.

The main result is that the retail-loss response is state-dependent. Retail long-side loss states are associated with more next-day negative discussion in lower-Neuroticism states, but the marginal response weakens when lagged board-day Neuroticism is already high, consistent with loss-salience saturation. The same state distinction also organizes subsequent outcomes: high-Neuroticism days are contemporaneously more negative but exhibit weaker next-day diffusion after conditioning on current negativity, and board-day Neuroticism also separates states with stronger versus weaker cumulative return association. The paper’s contribution is therefore to show that negative discussion matters not only by its level, but by whether it remains loss-responsive.

The rest of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the board data, the KRX flow data, and variable construction. Section 4 presents the empirical framework for retail-loss response, state dependence, discussion diffusion, and market

relevance. Section 5 reports the main mechanism results. Section 6 interprets board-day Neuroticism beyond scalar negativity, and Section 7 presents the market-side implications across saturation states. Section 8 concludes.

2 Related Literature

This paper connects to three lines of research—behavioral finance and textual sentiment, the social structure of investor communication, and psycholinguistic language measurement. Taken together, these literatures show that investor beliefs, attention, and social communication matter for markets, and they provide rich ways to measure sentiment and information transmission. But they typically characterize how negative, informative, or socially connected a discussion is, rather than whether the discussion still has marginal response capacity. Our contribution is to study this remaining-response margin directly: we ask when retail losses continue to generate negative board language and when loss-related discussion appears closer to saturation.

Behavioral-finance models establish that prices can reflect investor psychology and social interaction beyond fundamentals and risk. De Long et al. (1990) formalize how noise-trader beliefs affect prices under limits to arbitrage, Shiller (1995) analyzes conversation and information cascades as mechanisms behind herd behavior, and Hirshleifer (2001) surveys the role of investor psychology in asset pricing. The empirical sentiment literature operationalizes these ideas with observable proxies: Baker and Wurgler (2006) link aggregate sentiment to the cross-section of stock returns, and Da et al. (2015) use household search behavior to construct a market-level fear index that predicts short-term return reversals. These studies motivate treating investor psychology as empirically relevant, but they leave open how to observe the state of an ongoing retail discussion at the moment when losses are being processed.

Textual-finance research makes these behavioral forces observable. Early message-board work (Tumarkin and Whitelaw, 2001; Antweiler and Frank, 2004; Das and Chen, 2007) shows that retail-investor text contains information mixed with substantial noise. Tetlock (2007); Tetlock et al. (2008) demonstrate that pessimistic media language predicts low returns and subsequent reversion; García (2013) shows that sentiment predictability is stronger in recessions; Tetlock (2011) documents investor reactions to stale text; and Loughran and McDonald (2011, 2016) provide finance-specific dictionary infrastructure and implementation guidance. Social-media studies extend the analysis to retail platforms: Chen et al. (2014) show that stock opinions transmitted through social media predict future returns and earnings surprises, Sprenger et al. (2014) link microblog sentiment to returns, volume, and volatility, and Renault (2017) shows that intraday StockTwits sentiment forecasts S&P 500 closing returns. Across this literature, text is usually summarized as tone, sentiment, attention, disagreement, or an opinion index. Those measures are valuable for measuring the *level* of investor sentiment, but they do not by themselves distinguish an unsaturated negative state from a saturated one. This distinction is central here: among stock-days with comparable

current negativity, the paper asks whether board-day language composition helps identify whether additional losses still translate into further negative discussion. Answering that question requires moving from text as a scalar signal to text as part of a social transmission process.

Once the question shifts from sentiment level to remaining response capacity, the social architecture of investor communication becomes important. [Hong et al. \(2004\)](#) document the role of social interaction in stock-market participation, while [Barber and Odean \(2008\)](#), [Da et al. \(2011\)](#), and [Boulland et al. \(2017\)](#) establish that attention and salience shape retail-investor behavior. Recent platform-level analyses isolate structure missed by aggregate sentiment indices: [Cookson and Niessner \(2020\)](#) decompose disagreement into within- and across-approach components, [Cookson et al. \(2023\)](#) document selective exposure with trading consequences on StockTwits, and [Welch \(2022\)](#) provides complementary evidence on retail-investor responses during market stress. A separate strand studies the propagation of beliefs, ideas, or news through investor networks: [Rantala \(2019\)](#) examines idea spread in a Ponzi-scheme setting, [Han et al. \(2022\)](#) formalize a social-transmission mechanism in which active strategies disproportionately spread, [Hirshleifer et al. \(2025\)](#) show that social transmission of public news shapes the speed and persistence of market reactions, and [Sui and Wang \(2025\)](#) document self-enhancing transmission bias on an online investor platform. These studies emphasize social interaction, selective exposure, and information transmission. Our setting is complementary: without observing an explicit follower network, we ask whether the board-day *content mix* itself contains information about remaining diffusion potential. That approach makes the measurement problem sharper, because the relevant object is not only whether posts are negative but also what kind of language composition accompanies that negativity.

To use content mix as a state variable, the paper also draws on research linking language to psychological constructs. [Pennebaker et al. \(2003\)](#) and [Tausczik and Pennebaker \(2010\)](#) establish the LIWC tradition, in which transparent word categories are used to study psychological states, social context, and aspects of personality-relevant language. That tradition is conceptually adjacent to, but methodologically distinct from, this paper’s automated labeling of short investor-board messages. In finance and accounting, related applications usually focus on managerial language or on stable individual traits rather than on transient discussion states: [Larcker and Zakolyukina \(2012\)](#) use psychologically grounded language categories to study deception in conference calls, [Hrazdil et al. \(2020\)](#) use machine-learning-based executive personality measures in an audit setting, and [Jiang et al. \(2024\)](#) connect surveyed household Big Five traits to beliefs, preferences, social interaction tendencies, and portfolio allocation. These papers motivate the idea that psychologically meaningful language can matter in financial settings. The object here is different: board-day Neuroticism is a stock-board-day share of Neuroticism-like language, used to summarize the state of an ongoing discussion rather than the trait of an identified investor. The remaining question is whether that language state corresponds to an economically meaningful margin, rather than only to a different textual description of negativity.

The state interpretation becomes most useful when it is tied to a non-textual economic shock. Using KRX individual-investor flows and prices, the paper constructs position-implied retail long-side

losses and asks whether these losses generate negative board language differently across board-day Neuroticism states. The loss variable is built outside the text-labeling procedure, so it provides an external economic check on the language-state interpretation. The empirical pattern is consistent with loss-salience saturation: retail losses generate negative discussion, but the marginal language response is weaker when lagged board-day Neuroticism is already high. This result links the three literatures above: losses provide the behavioral shock, board discussion provides the social transmission setting, and Neuroticism-like language provides the state variable.

Accordingly, the paper’s contribution is specific. It extends textual finance from the level of negative sentiment to the marginal responsiveness of negative discussion to retail losses. It extends social-transmission work by showing that a board-day content mix can separate diffusion-active negative discussion from more saturated discussion even without observing a follower network. And it uses Neuroticism-like language as a compositional state variable, not as a stand-alone sentiment index. The subsequent diffusion and market tests then ask whether the same state distinction also appears in later negative-discussion dynamics and in the price relevance of current negative discussion.

3 Data and Measurement

3.1 Stock-board sample and retail-flow data

This section constructs the two empirical objects needed for the paper’s state-dependent loss-response design: retail long-side loss states and board-day Neuroticism. The first measures whether recently observed aggregate retail long lots are in loss territory; the second summarizes the language state of the board on the same stock-day.

The setting is the panel of Korean stock message boards, where retail discussion remains a primary venue for public market talk and where [Choe et al. \(2005\)](#) provide evidence that domestic investors are a non-trivially informed segment of the equity market. The unit of observation is the stock-day. The 99 boards in the panel are selected as the most actively posted-to retail boards over the sample window for which both message archives and matched market data are available; the baseline daily panel contains 36,881 stock-days across 942 trading dates. The stock-level reduced-form diffusion exercise is estimated separately for the 95 boards with enough valid consecutive observations to support a converging fit. Board posts are assigned to trading days using a 15:30 cut-off, so that post-close messages are shifted to the next trading day. The same alignment rule is used in the intraday and disclosure-control exercises.

We complement the board data with KRX individual-investor flow data matched one-to-one by stock and trading date. These data allow us to construct stock-day retail long-side loss states used in the mechanism exercises. A key design choice is to use a *long-only* construction: a signed cumulative net-flow measure would incorrectly read negative accumulated retail flow as if it were short exposure,

whereas the present application is about whether previously accumulated long-side retail positions become loss-bearing when prices fall. This treatment is also appropriate in the Korean setting, where retail investors have limited practical access to short selling; negative accumulated retail net buying is therefore better interpreted as the absence of observed long-side exposure than as a short position.

The archive begins with message-level labels for sentiment polarity and Big Five personality categories, produced by an automated text-labeling procedure applied to message text after standard cleaning. The present manuscript treats those labels as inputs and aggregates them to the stock-day level; we use the labels as board-day language measures rather than as measures of stable investor traits, and we assess their economic content through external flow-based validation and robustness exercises rather than by assuming a structural measurement model. Let $M_{i,t}$ denote the number of labeled messages for stock i on day t . The main outcome is the daily negative-share measure

$$Neg_{i,t} = \frac{\text{negative messages}_{i,t}}{M_{i,t}}.$$

The focal regressor is

$$BN_{i,t} = \frac{\text{Neuroticism-labeled messages}_{i,t}}{M_{i,t}},$$

which we refer to as board-day Neuroticism. In equations, tables, and the archived panel, this variable is denoted **bN**. The paper also uses the daily shares of Agreeableness, Conscientiousness, Extraversion, and Openness, denoted **bA**, **bC**, **bE**, and **bO**. Because these shares are compositional and sum to one, specifications that include the full Big Five vector omit one reference category; throughout the conservative daily-panel exercises, Agreeableness is the omitted share.

Appendix Table [A1](#) reports label diagnostics available from the redistributable stock-day panel. Negative messages are overwhelmingly Neuroticism-like, confirming that sentiment and Neuroticism-like labels are tightly linked inside negative posts. We therefore do not treat raw board-day Neuroticism as a sentiment-free Big Five measure. The empirical designs below instead ask whether board-day Neuroticism contains state information after conditioning on the current level of negative discussion and after validating the state interpretation against non-textual retail loss states. Neuroticism-like labels also appear among non-negative messages, but the paper does not attribute the main diffusion result to non-negative messages specifically.

To reduce measurement noise in share outcomes, the daily-panel sample keeps stock-days with at least ten messages. Attention is measured as $\log(1 + M_{i,t})$. Same-day market controls include raw return, absolute return, and realized volatility, together with one-day lags of return and volatility. For the disclosure-control exercise, the paper merges Korean Exchange (KRX KIND) disclosure information at the stock-day level.

The KRX investor-flow merge contains 35 source-level gaps on stock-days with no observed trading activity, zero returns, and zero realized volatility. We set retail flow to zero for these no-trade observations; excluding them after shock construction leaves the retail-loss mechanism estimates

unchanged.

3.2 Language, exposure, and market variables

Several derived variables are useful for interpretation. The variable **bn_neg** measures the share of Neuroticism-labeled messages among *negative* messages only. This is closer to a “style of negativity” measure than raw board-day Neuroticism, which is computed over all messages. The variable **clr_bn** is the centered-log-ratio transform of daily Neuroticism share within the Big Five composition. The paper also constructs decile bins of current negativity within each stock and uses them for within-bin comparisons that hold the current level of negativity approximately fixed. Because the Big Five shares sum to one, the coefficient on **bn** in any specification with the remaining shares as controls is identified relative to the omitted reference; we use Agreeableness as the reference throughout, and report the reference-invariant centered-log-ratio version (**clr_bn**) as a robustness check in Section 6.2.

Table 1 reports summary statistics for the baseline next-day-regression sample and the added KRX retail-flow variables. The average stock-day contains 87 messages, the mean current negative share is 0.390, and the mean board-day Neuroticism share is 0.448. The added flow variables show substantial daily retail trading activity and provide the basis for the rolling retail loss-state measures.

Table 1: Summary statistics: baseline daily-panel sample

Variable	Mean	Std. dev.	P25	Median	P75
Messages per day (M)	87.0042	173.2467	20.0000	40.0000	89.0000
Current negative share (Neg_t)	0.3902	0.1157	0.3143	0.3919	0.4686
Next-day negative share (Neg_{t+1})	0.3884	0.1172	0.3125	0.3916	0.4681
Daily Neuroticism share (bn)	0.4481	0.1104	0.3793	0.4553	0.5263
Daily Conscientiousness share	0.1478	0.0813	0.0833	0.1379	0.1944
Daily Openness share	0.1851	0.0745	0.1356	0.1818	0.2286
Attention: $\log(1 + M_t)$	3.8572	1.0003	3.0445	3.7136	4.4998
Return	-0.0003	0.0293	-0.0144	0.0000	0.0122
Absolute return	0.0196	0.0218	0.0055	0.0134	0.0264
Realized volatility	0.0009	0.0030	0.0000	0.0002	0.0007
Retail buy value (KRW bn)	39.7858	94.5168	6.2590	14.2849	38.0207
Retail sell value (KRW bn)	39.6948	95.6635	6.1693	14.4321	37.7524
Retail net buy value (KRW bn)	0.0910	42.9304	-4.2480	0.0460	4.1496
Cumulative net exposure (KRW bn)	131.9661	1118.8902	-121.2158	12.9474	174.9678
Cumulative gross exposure (KRW bn)	3208.9375	7837.8242	465.3887	1270.1405	2879.4740
Retail long exposure share	0.1140	0.1800	0.0000	0.0208	0.1638
Rolling 20-day retail loss state (pp)	0.0388	0.1323	0.0000	0.0052	0.0326
Rolling 20-day retail gain state (pp)	0.0261	0.1232	0.0002	0.0040	0.0199

Notes: Sample: 36,881 stock-days, 99 stocks, and 942 trading dates. Value variables are in billions of Korean won. Retail loss and gain states are shown as percentages of market capitalization and use nonmissing matched-flow observations; regressions use standardized states.

4 Empirical Framework

4.1 A simple framework for loss-salience saturation

The empirical design is guided by a simple state-variable framework. The purpose is not to estimate a structural model of investor preferences or a prospect-theory loss-aversion parameter. Instead, the framework organizes the paper’s three empirical links: retail losses generate negative discussion, the response depends on the board’s language state, and the same state distinction appears in subsequent diffusion and market relevance.

Let $n_{i,t}$ denote the negative-message share on stock board i at date t . Let $s_{i,t} \in [0, 1]$ denote a latent loss-salience saturation state: high $s_{i,t}$ means that loss-related attention has already been extensively expressed on the board, while $1 - s_{i,t}$ is the remaining capacity for additional losses to generate new negative discussion. The observed Neuroticism-like share is a noisy language proxy for this state,

$$BN_{i,t} = b_0 + b_1 s_{i,t} + u_{i,t}, \quad b_1 > 0. \quad (1)$$

In this framework, bN summarizes a board-day language state that is informative about remaining loss-salience capacity.

Let $L_{i,t} \geq 0$ and $G_{i,t} \geq 0$ denote retail long-side loss and gain states. A reduced-form loss-response equation is

$$\Delta n_{i,t+1} = \alpha + (\lambda_L L_{i,t} + \lambda_G G_{i,t}) (1 - s_{i,t-1}) - \delta n_{i,t} + \varepsilon_{i,t+1}, \quad \lambda_L > \lambda_G \geq 0. \quad (2)$$

The loss-gain asymmetry $\lambda_L > \lambda_G$ captures the idea that losses are more salient for negative discussion than comparable gains. The interaction with $1 - s_{i,t-1}$ captures saturation: when the board is already in a high-salience state, the marginal response to further losses is weaker. Differentiating Equation (2) gives

$$\frac{\partial \Delta n_{i,t+1}}{\partial L_{i,t}} = \lambda_L (1 - s_{i,t-1}), \quad \frac{\partial^2 \Delta n_{i,t+1}}{\partial L_{i,t} \partial s_{i,t-1}} = -\lambda_L < 0.$$

Using Equation (1), this maps directly into the empirical prediction that the coefficient on $\text{Loss}_{i,t} \times BN_{i,t-1}$ is negative.

The same state variable also separates current intensity from future diffusion. For example, if discussion evolves according to

$$n_{i,t+1} = n_{i,t} + \phi(1 - s_{i,t})n_{i,t}(1 - n_{i,t}) - \gamma n_{i,t} + \varepsilon_{i,t+1}, \quad (3)$$

then, holding current negativity fixed, higher saturation lowers subsequent diffusion:

$$\frac{\partial n_{i,t+1}}{\partial s_{i,t}} = -\phi n_{i,t}(1 - n_{i,t}) < 0.$$

Finally, if price relevance depends on diffusion-active negativity rather than on raw negativity alone, cumulative future returns can be summarized as

$$R_{i,t,H} = \sum_{h=1}^H r_{i,t+h} = -\mu_H n_{i,t}(1 - s_{i,t}) + \xi_{i,t,H}, \quad \mu_H > 0. \quad (4)$$

This implies that current negative discussion should have a stronger negative return association in low-bN states, where remaining salience capacity is high, than in high-bN states, where the response has already flattened. The framework therefore yields two direct discussion-side predictions: losses should move negative discussion more than gains, and the marginal loss response should decline with lagged bN. It also yields a market-relevance implication: if diffusion-active negativity is more price-relevant, current negative discussion should have a stronger cumulative return association in low-bN states.

4.2 Retail loss states and state-dependent language response

The mechanism exercises use stock-day retail long-side loss states constructed from individual-investor flows and prices. The primary specification uses a rolling 20-day lot construction. On days with positive individual-investor net buying, we create an aggregate retail long lot with quantity $q_{i,s}$ and entry price $p_{i,s}^e$. Subsequent negative net buying reduces active lots using a FIFO rule. Let $\mathcal{A}_{i,t}^{20}$ denote active observed retail long lots created during the previous 20 trading days. The rolling loss and gain states are

$$C\text{Loss}_{i,t}^{20} = \frac{\sum_{\ell \in \mathcal{A}_{i,t}^{20}} q_{\ell} \max(0, p_{\ell}^e - P_{i,t})}{MktCap_{i,t}}, \quad C\text{Gain}_{i,t}^{20} = \frac{\sum_{\ell \in \mathcal{A}_{i,t}^{20}} q_{\ell} \max(0, P_{i,t} - p_{\ell}^e)}{MktCap_{i,t}}.$$

Thus the loss variable measures the unrealized loss state of recently observed retail long lots, with each lot's gain or loss evaluated relative to its entry price. This long-only construction treats negative accumulated retail net buying as reductions in observed long lots rather than as short positions. Because pre-sample holdings are unobserved, the state variables are not measures of total retail holdings; they are proxies generated by observed retail trading during the sample period. To assess sensitivity to the loss-state definition, Appendix Table A6 repeats the analysis using anchored cumulative levels and daily deterioration measures. In the regression tables, the loss and gain states are winsorized at the 1st and 99th percentiles and standardized separately.

The first question is whether retail long-side losses generate more next-day negative discussion than comparable gains:

$$\Delta Neg_{i,t+1} = \alpha_i + \lambda_t + \beta^- C\text{Loss}_{i,t}^{20} + \beta^+ C\text{Gain}_{i,t}^{20} + \Gamma' X_{i,t} + \varepsilon_{i,t+1}, \quad (5)$$

where $\Delta Neg_{i,t+1} = Neg_{i,t+1} - Neg_{i,t}$, α_i and λ_t are stock and date fixed effects, and $X_{i,t}$ contains current negative share, attention, same-day return, same-day absolute return, same-day realized

volatility, lagged return, and lagged realized volatility. The basic prediction is $\beta^- > \beta^+$.

The second question is whether this loss-state response is weaker when the board is already in a high-bN state:

$$\begin{aligned} \Delta Neg_{i,t+1} = & \alpha_i + \lambda_t + \beta^- CLoss_{i,t}^{20} + \beta^+ CGain_{i,t}^{20} \\ & + \theta \left(CLoss_{i,t}^{20} \times BN_{i,t-1} \right) + \delta BN_{i,t-1} + \Gamma' X_{i,t} + \varepsilon_{i,t+1}, \end{aligned} \quad (6)$$

with prediction $\theta < 0$. Under Equation (6), the marginal effect of the retail loss state is

$$\frac{\partial \Delta Neg_{i,t+1}}{\partial CLoss_{i,t}^{20}} = \beta^- + \theta BN_{i,t-1},$$

so a negative θ implies that the response to retail losses weakens as lagged bN rises. We interpret this pattern as reduced-form evidence of loss-salience saturation rather than as a structural estimate of investor loss aversion. All retail-loss regressions are weighted by next-day message count, or by current-day message count when the next-day count is unavailable, and standard errors are clustered two ways by stock and date.

4.3 Discussion diffusion specification

The stock-level exercise uses a parsimonious diffusion equation to summarize nonlinear propagation in negative-message share:

$$n_{i,t+1} = n_{i,t} + \beta_{i,t} n_{i,t} (1 - n_{i,t}) - \gamma_i n_{i,t}, \quad (7)$$

with

$$\log(\beta_{i,t}) = \beta_{0,i} + \theta' X_{i,t}. \quad (8)$$

Here $n_{i,t}$ is daily negative share rather than a literal infection rate. Silent readers are unobserved, the active set of writers changes over time, and the susceptible population is not fixed. Accordingly, Equations (7)–(8) are interpreted as a compact reduced-form description of diffusion rather than as a structural epidemic model. The stock-level coefficients are used primarily for sign stability, not for structural calibration.

4.4 Daily-panel implementation

The daily-panel diffusion analysis uses two related specifications. The baseline parametric specification relates next-day negativity to current-day language composition:

$$Neg_{i,t+1} = \alpha_i + \lambda_t + \rho Neg_{i,t} + \theta BN_{i,t} + \eta (BN_{i,t} \times Neg_{i,t}) + \Gamma' Z_{i,t} + \varepsilon_{i,t+1}, \quad (9)$$

where α_i and λ_t are stock and date fixed effects, respectively, and $Z_{i,t}$ contains the remaining Big Five shares, attention, same-day return, same-day absolute return, same-day realized volatility, lagged return, and lagged volatility. Regressions are weighted by next-day message count so that stock-days with more precisely measured shares receive more weight.

The preferred diffusion design replaces the parametric treatment of current negativity with within-bin comparisons that condition on stock-specific decile bins of $Neg_{i,t}$:

$$Neg_{i,t+1} = \alpha_i + \lambda_t + \theta BN_{i,t} + \Pi_{i,b(Neg_{i,t})} + \Gamma'W_{i,t} + \varepsilon_{i,t+1}, \quad (10)$$

where $\Pi_{i,b(Neg_{i,t})}$ denotes stock-specific current-negativity-bin fixed effects and $W_{i,t}$ contains the reference-coded Big Five composition, attention, market controls, and, in the richest specification, trajectory and disclosure controls. The design asks whether, among days with similar current negativity, higher **bN** predicts more or less next-day diffusion. The within-bin design is preferred over a parametric absorption of the full Big Five composition because the compositional shares are highly correlated; parametric absorption loses precision, while stock-specific decile bins absorb the level of current negativity nonparametrically without that precision loss (the parametric specification is reported as a robustness check in Appendix Table A10). Standard errors are clustered two ways by stock and date in the within-bin specifications.

We treat the regressions as descriptive rather than causal: the identifying variation comes from within-stock changes in board composition after removing common date effects and controlling for current negativity and market conditions, and we interpret the coefficients as reduced-form discussion-state coefficients.

5 Retail Losses and Loss-Salience Saturation

This section reports the paper’s main mechanism evidence. The question is whether retail long-side loss states generate negative discussion, and whether that marginal language response depends on the lagged **bN** state of the board. The results show this pattern: retail loss states increase next-day negative discussion more than comparable gain states, but the marginal response is weaker when lagged **bN** is already high.

Table 2 reports the main estimates using the rolling 20-day loss-state measure from Section 4. Column (1) shows the direct loss/gain response. Column (2) shows that the response is state-dependent: the interaction between the retail loss state and lagged **bN** is -0.0157 ($p = 0.003$). The implied response to an interquartile increase in the retail loss state declines across the lagged-**bN** distribution and is close to zero in the upper quartile. Appendix Tables A4 and A5 show that this attenuation is not driven by raw return components or by the linear share specification.

Figure 1 visualizes the mechanism. The fitted surface is steepest when the retail loss state is high and lagged **bN** is low, and it flattens as lagged **bN** rises. The marginal-effect panel shows the same

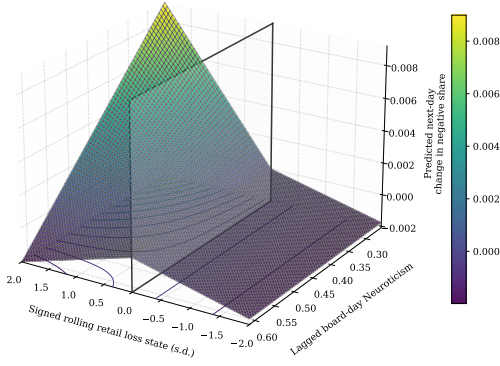
Table 2: Retail losses and loss-salience saturation

	(1) Loss/gain states	(2) State-dependent response
Retail loss state	0.0008 (0.0005)	0.0086*** (0.0026)
Retail gain state	-0.0011** (0.0005)	-0.0009* (0.0005)
Lagged bN		0.0933*** (0.0107)
Loss state $\times$ lagged bN		-0.0157*** (0.0051)
IQR loss-state effect at P25 lagged bN		0.0007*** (0.0002)
IQR loss-state effect at median lagged bN		0.0004*** (0.0001)
IQR loss-state effect at P75 lagged bN		0.0001 (0.0001)
Window	20 trading days	20 trading days
Stock fixed effects	Yes	Yes
Date fixed effects	Yes	Yes
Controls	Yes	Yes
Count weights	Yes	Yes
Observations	33,868	33,868

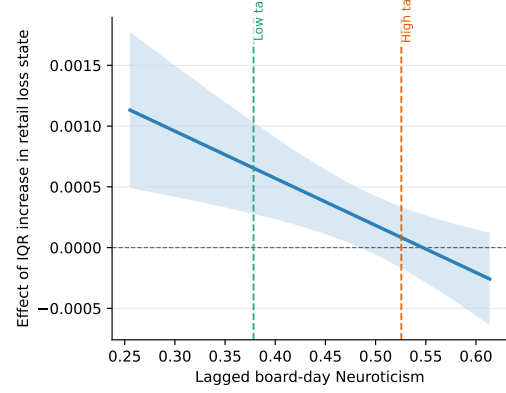
Notes: Dependent variable: $\Delta Neg_{i,t+1} = Neg_{i,t+1} - Neg_{i,t}$. Loss and gain variables are standardized rolling 20-day retail unrealized loss and gain states. The P25, median, and P75 rows report the effect of an interquartile increase in the retail loss state at the corresponding lagged-**bN** percentiles. Controls include current negative share, attention, return, absolute return, realized volatility, and lagged return and volatility. Standard errors are clustered by stock and date. $\tilde{p}^{**} < 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

pattern directly: as the board enters a higher-**bN** state, the language response to an interquartile increase in the retail loss state moves from a steeper to a flatter region of the response curve.

Appendix Table A6 compares the main rolling 20-day loss-state definition with anchored cumulative levels and daily deterioration measures. The interaction between retail losses and lagged **bN** remains negative and statistically significant across these alternatives.



Panel A. Response surface



Panel B. Marginal loss sensitivity

Figure 1: Retail-loss response surface and marginal loss sensitivity

Notes: Panel A plots fitted ΔNeg_{t+1} over signed standardized rolling retail gain/loss states and lagged bN ; negative x-values denote the gain-state domain and positive x-values denote the loss-state domain. Panel B plots the effect of an interquartile increase in the rolling retail loss state, $(\beta^- + \theta BN_{i,t-1})$ scaled by the loss-state interquartile range.

6 Interpreting b_N Beyond Scalar Negativity

The loss-response evidence implies that bN should not be read as another scalar measure of negativity. If high- bN states are saturated states, they should be negative in the present but less diffusion-active afterward. This section tests that implication directly by asking whether board-day bN predicts future negativity after the current level of negative discussion is held fixed.

Table 3 reports the main daily-panel estimates. Column (1) is the baseline count-weighted specification, and columns (2) and (3) report the within-bin specification of Section 4. The coefficient on bN is negative in all three columns and remains statistically significant after stock-specific current-negativity-bin fixed effects are added. In the preferred within-bin estimate, the coefficient is -0.0874 and significant at the 1% level.

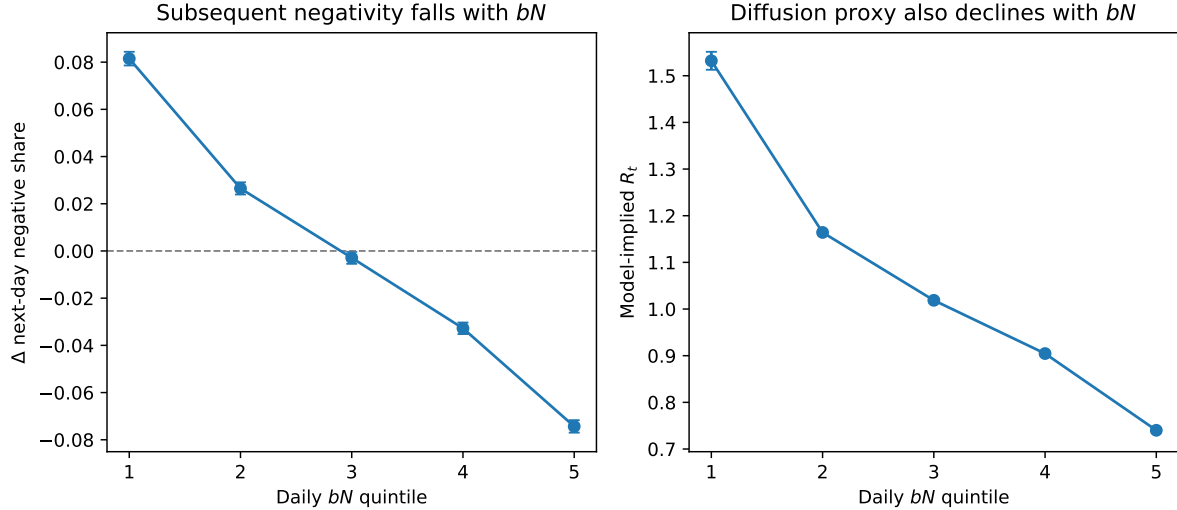
These estimates support the saturation-state interpretation from Section 5. Higher daily bN composition predicts *less* subsequent diffusion of negative sentiment, and the strongest evidence comes from the within-bin comparisons in columns (2) and (3), where the current level of negativity is held fixed directly. Appendix Table A9 shows that the result is not absorbed by one-day reversals in negative discussion, and Appendix Table A11 reports consistent board-level sign patterns.

Figure 2 provides a nonparametric summary of the same pattern: moving from the lowest to the highest bN quintile, the average next-day change in negative share declines from strongly positive to strongly negative, and the model-implied diffusion proxy declines as well.

Table 3: Daily-panel evidence on future negativity

	(1) Baseline FE	(2) Within-bin	(3) Within-bin + trajectory
Daily bN	−0.1535** (0.0632)	−0.0874*** (0.0281)	−0.0641** (0.0251)
bN × current neg.	0.2052* (0.1074)		
Current-negativity bins	No	Yes	Yes
Trajectory controls	No	No	Yes
Disclosure controls	No	No	Yes
Other Big Five shares	Partial	Full, ref.-coded	Full, ref.-coded
Stock fixed effects	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes
Clustering	Stock	Stock + date	Stock + date
Observations	36,881	36,877	36,698
Stocks	99	98	98

Notes: Dependent variable: $Neg_{i,t+1}$. Columns (2)–(3) include stock-specific current-negativity-bin fixed effects. All columns are count-weighted by next-day messages. Standard errors are clustered by stock in (1) and by stock and date in (2)–(3). $\hat{p}^{**} < 0.01$, $\hat{p}^* < 0.05$, $\hat{p} < 0.10$.

**Figure 2:** Diffusion metrics by daily bN quintile

Notes: Left: mean next-day change in negative share by daily bN quintile with 95% CIs. Right: corresponding model-implied diffusion proxy.

6.1 Intensity versus diffusion

The same-day and next-day relationships between **bN** and negative share move in opposite directions. Same-day, the two are tightly linked: in a regression of current-day negative share on current-day **bN** with stock and date fixed effects, the coefficient is 0.7671 and statistically significant at any conventional level (Table 4, Panel A). High-**bN** days are unmistakably high-negativity days. Conditional on that contemporaneous association, however, higher **bN** predicts *less* subsequent negativity, with within-bin coefficients ranging from -0.087 to -0.064 depending on the controls (Table 3, columns 2–3).

This contrast is the empirical content of the intensity-versus-diffusion distinction. The relevant object is conditional board-day language composition: after absorbing stock fixed effects, date fixed effects, and stock-specific current-negativity bins, higher **bN** predicts weaker subsequent diffusion. Appendix Table A3 reports the equivalent residualized-**bN** representation and component diagnostics.

6.2 Conditional composition diagnostics

Table 4 keeps the core interpretation visible in the text. Column (1) shows that same-day **bN** is strongly associated with current negative share. Column (2) then reports the preferred within-bin next-day estimate from Table 3: after conditioning on current negativity, higher **bN** predicts less subsequent negative discussion. Column (3) adds the intraday diagnostic that morning **bN** does not mechanically lower afternoon negativity. Appendix Table A2 reports the remaining composition and disclosure checks.

Table 4: Current intensity versus subsequent diffusion

	(1) Current neg.	(2) Next-day neg.	(3) Afternoon neg.
Daily bN	0.7671*** (0.0301)	-0.0874^{***} (0.0281)	
Morning bN			0.0234 (0.0302)
Current-negativity bins	No	Yes	No
Stock fixed effects	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes
Observations	36,881	36,877	26,168

Notes: Column (1) uses current negative share, column (2) next-day negative share, and column (3) afternoon negative share. Standard errors are clustered by stock and date where applicable. $\tilde{p}^* \leq 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

7 Market Relevance Across Saturation States

This section asks whether the same loss-response state distinction appears in the price relevance of current negative discussion. The market implication is conditional: board-day Neuroticism should

matter because it separates negative discussion that remains loss-responsive and diffusion-active from negative discussion that is closer to saturation. We first document the direct lead-lag relation between **bN** and market conditions, then show that the cumulative return association with current negativity is concentrated in low-**bN** discussion states, and finally connect the pattern to attention exhaustion.

7.1 Unconditional market lead-lag

The lead-lag relationship between **bN** and market outcomes is informative for interpreting what **bN** measures. Table A13 shows that daily **bN** does not directly forecast future returns, absolute returns, or $\log(1+RV)$ at horizons one through three.

The reverse direction is stronger. Table 5 regresses next-day **bN** and next-day **neg_share** on current market variables, controlling for current **bN**, current **neg_share**, attention, and stock and date fixed effects. Lower daily returns and higher realized volatility predict higher next-day **bN**, with all three return- and volatility-related coefficients statistically significant at the 5% level or better; the same pattern holds for next-day **neg_share**. Thus, market conditions help shape the high-**bN** discussion state, while **bN** itself is not a direct short-horizon market forecast.

Table 5: Market feedback: current market conditions and next-day board language

	(1) Next-day bN	(2) Next-day neg_share
Daily return	−0.0949*** (0.0266)	−0.1033*** (0.0295)
Absolute return	−0.1709** (0.0661)	−0.2092*** (0.0745)
Realized volatility	0.9202** (0.3667)	0.9473*** (0.3505)
Current bN , neg_share , attn	Yes	Yes
Stock fixed effects	Yes	Yes
Date fixed effects	Yes	Yes
Clustering	Stock + date	Stock + date
Observations	37,763	37,763

Notes: Dependent variables are measured at $t + 1$; regressors are current-day market variables. Controls include current **bN**, **neg_share**, attention, stock fixed effects, and date fixed effects. Standard errors are clustered by stock and date. $\hat{p}^{**} < 0.01$, $\hat{p}^* < 0.05$, $\hat{p} < 0.10$.

7.2 State-conditional return association

Section 7.1 showed that the relevant market test is conditional rather than unconditional. If **bN** helps distinguish negative discussion that is still loss-responsive and diffusion-active from discussion already closer to saturation, the same distinction should also appear in how current negativity relates to future prices.

Our preferred specification uses within-stock quartile tails, which makes the language of “low-bN” and “high-bN” literal rather than median-based: low-bN days are those in the bottom quartile of a stock’s own bN distribution, and high-bN days are those in the top quartile. The middle 50% of observations are set aside so that the market comparison focuses on the clearest low- and high-bN states. We then run a state-conditional local projection of cumulative future returns on current negative share, allowing the slope on `neg_share` to differ across the low- and high-bN tails:

$$\sum_{h=1}^H ret_{i,t+h} = \alpha_i + \lambda_t + \beta_L Neg_{i,t} L_{i,t} + \beta_H Neg_{i,t} H_{i,t} + \gamma H_{i,t} + \Phi' W_{i,t} + \varepsilon_{i,t+H}, \quad (11)$$

where $L_{i,t}$ and $H_{i,t}$ indicate membership in the within-stock bottom and top quartiles of bN, respectively, $W_{i,t}$ contains current-day market controls and ten own-return lags as the local-projection convention, and standard errors are two-way clustered by stock and date.

Figure 3 and Table A14 report the quartile-tail cumulative estimates at horizons $H = 1, \dots, 20$. The low-bN-tail response is negative through the short-to-medium horizons and statistically significant through $H = 6$, reaching about -6.8pp at the one-week horizon. The high-bN-tail response remains close to zero over the same horizons. The implied cross-state difference $\beta_H - \beta_L$ is positive throughout the horizon path and statistically significant from $H = 2$ through $H = 6$. In other words, current-day negative content predicts cumulative return declines primarily in the low-bN tail, whereas the corresponding high-bN tail estimates remain muted.

The pattern is economically reminiscent of the partial-reversion trajectory documented for retail-driven sentiment in Tetlock (2011): the low-bN-tail response deepens over the first week and then only partly retraces. We read these estimates as state-dependent price relevance rather than as responses to exogenous shocks. Appendix Table A15 and Figures A1–A3 report smooth-transition and median-split alternatives.

This finding extends the paper’s loss-response framing from board dynamics to market dynamics. Low-bN states are the states in which negative discussion remains more responsive to losses and continues into subsequent discussion; they are also the states in which the return association with current negativity is most visible. High-bN states, by contrast, remain intense on impact but are much less associated with subsequent cumulative return drift. bN therefore appears to indicate whether a day’s negativity remains price-relevant because it is still diffusion-active, or whether it is already closer to saturation. The appendix shows that this directional return pattern is not mirrored in absolute return or realized volatility, which points to price relevance rather than a broad volatility response.

7.3 Attention exhaustion and remaining diffusion potential

The state-conditional results admit a coherent interpretation: conditional on the current level of negativity, bN indicates the *remaining diffusion potential* of an adverse discussion state. High-bN

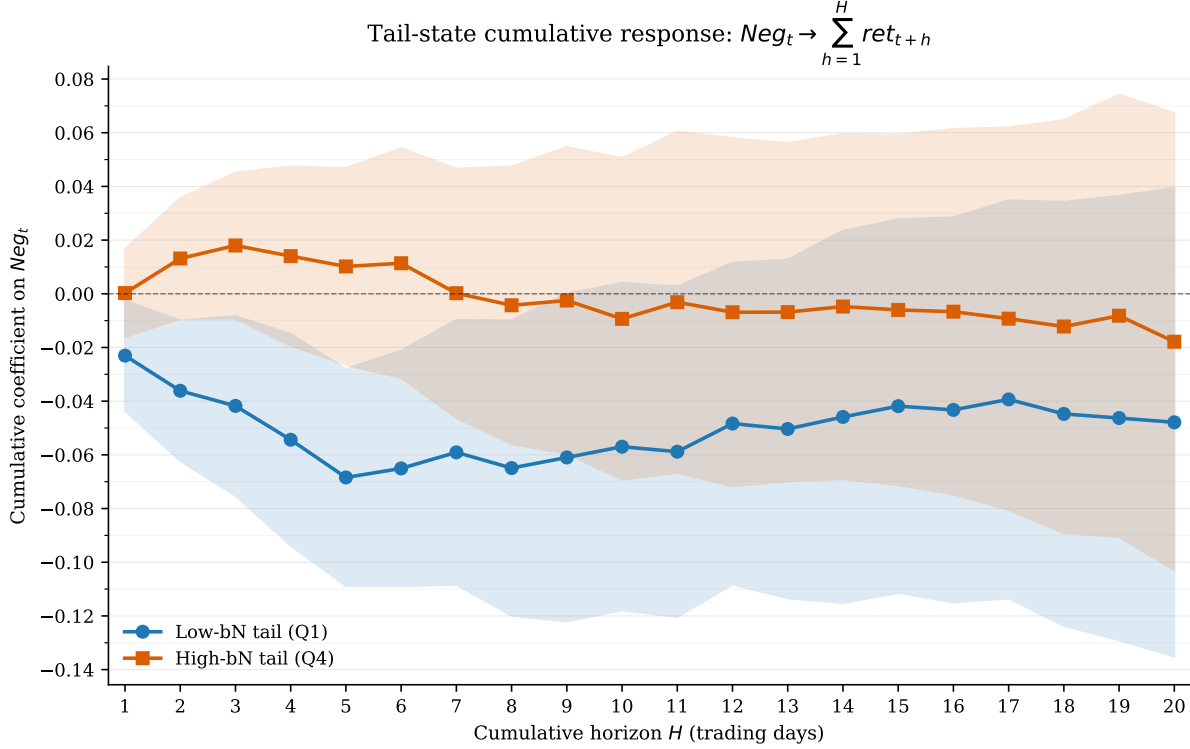


Figure 3: Tail-state market response to current-day negative share

Notes: The figure plots β_L and β_H from Equation (11) for cumulative returns $H = 1, \dots, 20$. Low/high states are within-stock bottom/top bN quartiles. Shaded bands are 95% confidence intervals.

days are days on which the stock’s retail attention appears already close to a peak. The same-day association between bN and current negativity is large and positive (Section 6); the next-day negativity is lower after conditioning on current negativity (the diffusion finding); and posting volume itself drops sharply over the following days, with the effect strengthening over the next few horizons (Appendix Table A12). On this reading, high-bN days are attention-saturated: the negative discussion has already concentrated investor attention on the stock today, and that attention is depleted by the following day rather than carried forward.

Two implications follow. First, high-bN negativity appears to have limited remaining momentum. If price relevance requires continued attention, discussion, or trading pressure, an attention-exhausted negative discussion state should have weaker subsequent price implications. Second, low-bN negativity looks different. The discussion has not yet exhausted its audience; attention and negative content still have room to diffuse; and the cumulative return response to current negative share is correspondingly stronger over the following trading days. The cumulative-return asymmetry documented in Section 7.2 is the market-side counterpart of this filter.

We therefore interpret bN as a behavioral state variable only after conditioning on current negativity: it helps separate negative-discussion shocks with remaining diffusion potential—*diffusion-active* states—from those whose attention momentum has already been substantially depleted—*attention-*

exhausted states. The evidence is consistent with the former exhibiting a stronger subsequent price response. The key implication for sentiment measurement is that aggregate scalar negativity measures, by themselves, cannot distinguish live shocks from spent shocks; conditional **bN** variation provides a language-based indicator for that distinction.

8 Conclusion

This paper shows that online investor discussion is not fully described by the level of negativity. What matters is also whether negative discussion remains responsive to retail losses. Using KRX individual-investor flows and stock prices, we construct stock-day retail long-side loss-state measures and show that the marginal language response to retail loss states weakens when lagged board-day Neuroticism is already high. This pattern supports interpreting board-day Neuroticism as a conditional language-state variable for retail loss-salience saturation.

The remaining results reinforce this interpretation. High-Neuroticism days are intensely negative in the present but self-limiting in subsequent discussion, while lower-Neuroticism states are the states in which current negativity has a stronger cumulative return association. The same language state therefore organizes both discussion dynamics and the market relevance of negative discussion.

The remaining step is to identify the market-transmission channel more directly. The present evidence shows that board-day Neuroticism separates loss-responsive, diffusion-active negative discussion from more saturated negative discussion, and that this distinction is also visible in cumulative return associations. It does not, however, isolate the trading, information, or network mechanism through which diffusion-active discussion becomes market relevant. Distinguishing among these channels requires data that link language states more directly to investor identities, information arrival, or order-book conditions. This is the natural next step for understanding when retail discussion remains merely expressive and when it becomes part of price formation.

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A Appendix

A.1 LLM labeling prompt (sentiment and Big Five)

For reproducibility, the labeling pipeline uses the following base prompt template, with each forum comment inserted at `<COMMENT_TEXT>`.

You are an expert in analyzing stock market discussions and psychological traits. The given content is a comment from a stock market forum. Your task is twofold: first, extract the psychological sentiment of this comment as Positive, Negative, or Neutral. Second, identify the most relevant Big Five personality trait (Openness to Experience, Conscientiousness, Extraversion, Agreeableness, Neuroticism) based on the language and sentiment expressed in the comment.

"""`<COMMENT_TEXT>`"""

1) Based on the overall sentiment and language used in the text, assign one of the sentiment labels (Positive, Negative, Neutral) that best represents the sentiment of the comment. Explain in detail why you chose this label by providing specific examples or phrases from the text that illustrate the sentiment.

2) Based on the sentiment and language style, assign one of the Big Five personality trait labels (Openness to Experience, Conscientiousness, Extraversion, Agreeableness, Neuroticism) that best represents the personality trait expressed in the comment. Provide specific examples or phrases from the text that support this classification.

Your response must be in JSON format with the following keys:

- Sentiment Label (Positive, Negative, Neutral)
- Sentiment Explanation
- Big Five Label (Openness to Experience, Conscientiousness, Extraversion, Agreeableness, Neuroticism)
- Big Five Explanation

A.2 Measurement and composition diagnostics

This subsection collects diagnostics for interpreting board-day Neuroticism as a conditional language-state variable. The tables document the relation between sentiment and Neuroticism-like labels, report alternative composition checks, and show the residualized-**bN** representation of the within-bin diffusion design.

Table A1: Label diagnostics for Neuroticism-like language

<i>Panel A. Neuroticism-like labels by sentiment domain</i>				
	Neuroticism-like	Other Big Five	Total messages	Neuroticism-like share
Negative messages	1,349,581	34,643	1,384,224	0.975
Non-negative messages	282,003	1,611,861	1,893,864	0.149
<i>Panel B. Stock-day correlations</i>				
	bN	bN_neg	bN_{nonneg}	<i>Neg</i>
bN	1.000	0.195	0.421	0.783
bN_neg	0.195	1.000	-0.052	-0.006
bN_{nonneg}	0.421	-0.052	1.000	-0.096
<i>Neg</i>	0.783	-0.006	-0.096	1.000

Notes: Panel A uses the binary negative/non-negative split available in the stock-day panel. Panel B reports stock-day correlations among **bN**, **bN_neg**, non-negative-message **bN**, and negative share.

Table A2: Additional diagnostics for interpreting b_N

<i>Panel A. Alternative composition and intraday diagnostics</i>				
	(1)	(2)	(3)	(4)
	Next-day neg.	Next-day neg.	Afternoon neg.	Afternoon neg.
bN_neg	0.0994** (0.0461)			0.0664*** (0.0214)
clr_bN		-0.0164 (0.0115)		
Morning bN			0.0234 (0.0302)	
Stock fixed effects	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes
Observations	36,516	36,881	26,168	26,168

<i>Panel B. Disclosure controls, decomposition, and grouped-binomial robustness</i>				
	(1)	(2)	(3)	
	Disclosure controls	Decomposition	Grouped binomial	
Daily bN	-0.1029* (0.0593)		-0.3533 (0.2471)	
Adverse disclosure	-0.0266** (0.0113)			
Negative-post component		-0.0023 (0.0490)		
Non-negative-post component		-0.0918** (0.0370)		
Difference: negative – non-negative		0.0895 (0.0730)		
Stock fixed effects		Yes	Yes	Yes
Date fixed effects		Yes	Yes	Yes
Observations		36,881	36,700	36,700

Notes: Each column reports a separate regression. Panel A reports alternative composition and intraday diagnostics; Panel B reports disclosure, decomposition, and grouped-binomial checks. Standard errors are clustered by stock and date where applicable. $\tilde{p}^{**} < 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

Table A3: Conditional b_N variation and component diagnostics

	(1)	(2)	(3)	(4)	(5)
bN	−0.0874*** (0.0281)				
Residualized bN		−0.0874*** (0.0281)			
Non-negative-message bN			0.0065 (0.0207)	0.0143 (0.0177)	0.0091 (0.0162)
Negative-message bN				0.0256 (0.0210)	0.0210 (0.0180)
ΔNeg_t					−0.1525*** (0.0090)
Stock fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Current-negativity bins	Yes	Yes	Yes	Yes	Yes
Count weights	Yes	Yes	Yes	Yes	Yes
Observations	36,877	36,877	36,871	36,506	36,420

Notes: Dependent variable: $\Delta Neg_{i,t+1}$. Column (2) uses **bN** residualized on stock fixed effects, date fixed effects, and stock-specific current-negativity bins. Columns (3)–(5) report component diagnostics. All columns use the within-bin diffusion controls; standard errors are clustered by stock and date. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.3 Retail-loss mechanism robustness

This subsection reports robustness checks for the retail-loss mechanism in Table 2. The checks compare the rolling 20-day retail loss-state measure with raw return components, estimate share-outcome models, vary the loss-state definition, examine next-day changes in **bN**, and replace lagged **bN** with other Big Five-like language shares.

Table A4: Retail loss states versus raw return benchmarks

	(1)	(2)	(3)	(4)
Down return	−0.0019 (0.0015)		−0.0028* (0.0017)	−0.0023 (0.0017)
Up return	−0.0061*** (0.0016)		−0.0082*** (0.0015)	−0.0081*** (0.0015)
Retail loss state		0.0013** (0.0006)	0.0008 (0.0005)	0.0086*** (0.0026)
Retail gain state		−0.0012** (0.0005)	−0.0011** (0.0005)	−0.0009* (0.0005)
Loss state × lagged bN				−0.0157*** (0.0051)
Lagged bN				0.0933*** (0.0107)
Stock fixed effects	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes
Count weights	Yes	Yes	Yes	Yes
Observations	36,881	33,868	33,868	33,868

Notes: Dependent variable: $\Delta Neg_{i,t+1}$. Down/up returns are $\max(0, -r_{i,t})$ and $\max(0, r_{i,t})$. Retail loss and gain are standardized rolling 20-day retail unrealized loss and gain states. Controls, fixed effects, weights, and clustering follow Table 2. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$.

Table A5: Share-outcome robustness for the retail-loss response

	(1) Fractional logit	(2) Grouped binomial
Retail loss state	0.0328*** (0.0112)	0.0380*** (0.0112)
Retail gain state	-0.0034 (0.0021)	-0.0036* (0.0021)
Lagged bN	0.3589*** (0.0470)	0.3870*** (0.0459)
Loss state $\times$ lagged bN	-0.0614*** (0.0233)	-0.0695*** (0.0219)
Outcome	Neg_{t+1}	Neg_{t+1}
Link	Logit	Logit
Frequency weights	M_t	M_{t+1}
Stock fixed effects	Yes	Yes
Date fixed effects	Yes	Yes
Controls	Yes	Yes
Clustering	Stock + date	Stock + date
Observations	33,868	33,868

Notes: Dependent variable: $Neg_{i,t+1}$. Column (1) is fractional logit with M_t frequency weights; column (2) is grouped binomial with M_{t+1} frequency weights. Coefficients are logit-scale estimates; standard errors are clustered by stock and date. $\tilde{p}^{**} < 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

Table A6: Alternative retail loss-state definitions

Definition	Loss/gain state	Loss state $\times$ bN	Obs.
Rolling 20-day level	0.0086***	-0.0157***	33,868
Anchored cumulative level	0.0087***	-0.0179***	36,792
Rolling 20-day deterioration	0.0062**	-0.0125**	32,835
Anchored deterioration	0.0068**	-0.0136***	36,792
Stock and date fixed effects	Yes	Yes	
Controls and count weights	Yes	Yes	

Notes: Dependent variable: $\Delta Neg_{i,t+1}$. The table reports the loss-state coefficient and its interaction with lagged **bN** across alternative lot-state definitions. Rolling levels are computed from active lots created in the previous 20 trading days; anchored levels use observed active lots accumulated from the start of the matched sample; deterioration measures use daily positive changes in the corresponding loss state. Controls, fixed effects, weights, and clustering follow Table 2. $\tilde{p}^{**} < 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

Table A7: Does the retail loss state directly move Neuroticism-like language?

	(1) Simple	(2) State-dependent
Retail loss state	0.0009** (0.0004)	0.0081*** (0.0023)
Retail gain state	−0.0006 (0.0004)	−0.0004 (0.0004)
Lagged bN		0.0640*** (0.0094)
Loss state × lagged bN		−0.0145*** (0.0046)
Stock fixed effects	Yes	Yes
Date fixed effects	Yes	Yes
Controls	Yes	Yes
Count weights	Yes	Yes
Observations	34,663	34,663

Notes: Dependent variable: $\Delta b_{N,i,t+1}$. The table repeats the rolling 20-day retail loss-state specification with next-day change in bN as the outcome. Controls, fixed effects, weights, and clustering follow Table 2. $\tilde{p}^{**} < 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

Table A8: Big Five placebo interactions in the retail-loss response

	(1) bN	(2) b_A	(3) b_C	(4) b_E	(5) b_O
Retail loss state	0.0086*** (0.0026)	−0.0011 (0.0011)	0.0030** (0.0014)	−0.0016 (0.0013)	−0.0012 (0.0019)
Retail gain state	−0.0009* (0.0005)	−0.0011** (0.0005)	−0.0010* (0.0005)	−0.0010** (0.0005)	−0.0011** (0.0005)
Loss state × trait	−0.0157*** (0.0051)	0.0433* (0.0257)	−0.0134* (0.0080)	0.0190* (0.0110)	0.0116 (0.0090)
Stock fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Count weights	Yes	Yes	Yes	Yes	Yes
Observations	33,868	33,868	33,868	33,868	33,868

Notes: Dependent variable: $\Delta Neg_{i,t+1}$. Each column replaces lagged bN with the indicated lagged Big Five-like language share in the rolling 20-day loss-state specification. Controls, fixed effects, weights, and clustering follow Table 2. $\tilde{p}^{**} < 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

A.4 Diffusion robustness

This subsection reports auxiliary diffusion checks. The tables add a one-day negativity-change control, vary weighting and horizons, and summarize board-level reduced-form diffusion fits.

Table A9: Within-bin diffusion and one-day negativity reversal

	(1)	(2)
bN	−0.0874*** (0.0281)	−0.0734*** (0.0260)
ΔNeg_t		−0.1536*** (0.0091)
Stock fixed effects	Yes	Yes
Date fixed effects	Yes	Yes
Count weights	Yes	Yes
Observations	36,877	36,789

Notes: Dependent variable: $Neg_{i,t+1}$. Column (2) adds ΔNeg_t . Both columns include stock, date, and stock-specific current-negativity-bin fixed effects and are count-weighted by next-day messages. Standard errors are clustered by stock and date. $\tilde{p}^* \leq 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

Table A10: Weighting and horizon robustness

	(1) Full comp.	(2) Unweighted	(3) Sqrt wt.	(4) $M_t \geq 30$	(5) $h = 2$	(6) $h = 3$
Daily bN	−0.1037* (0.0594)	−0.0489 (0.0320)	−0.0795* (0.0472)	−0.0706 (0.0680)	−0.0681 (0.0659)	−0.1067 (0.0713)
Outcome	Neg_{t+1}	Neg_{t+1}	Neg_{t+1}	Neg_{t+1}	Neg_{t+2}	Neg_{t+3}
Stock fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Full Big Five controls	Yes	Yes	Yes	Yes	Yes	Yes
Count weights	Yes	No	Sqrt	Yes	Yes	Yes

Notes: Each column reports the coefficient on daily **bN**. Columns (1)–(4) use Neg_{t+1} ; columns (5)–(6) use Neg_{t+2} and Neg_{t+3} . Standard errors are clustered by stock and date. $\tilde{p}^* \leq 0.01$, $\tilde{p}^* < 0.05$, $\tilde{p} < 0.10$.

Table A11: Stock-level reduced-form diffusion summaries

Specification	Mean loading	Median loading	Sig. negative	Sig. positive
Baseline stock-level diffusion fit: bN	−1.2646	−0.5475	28	3
Reference-coded diffusion fit: bN	−0.6248	−0.7029	20	2

Notes: Rows summarize 95 board-specific reduced-form diffusion fits. The reference-coded row omits Agreeableness. “Sig. negative/positive” counts estimates significant at the 5% level.

A.5 Attention robustness

This subsection reports the posting-volume diagnostic used to interpret high-**bN** days as attention-saturated states.

Table A12: Attention exhaustion: next-day posting volume on daily **bN**

	$h = 1$ $\log(1+M_{t+1})$	$h = 2$ $\log(1+M_{t+2})$	$h = 3$ $\log(1+M_{t+3})$
Daily bN	-0.281*** (0.090)	-0.379*** (0.115)	-0.424*** (0.113)
Observations	36,780	36,682	36,584
Stock fixed effects	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes
Clustering	Stock + date	Stock + date	Stock + date

Notes: Dependent variables are future $\log(1 + M)$ at horizon h . Controls include current attention, market controls, and ten own-return lags, with stock and date fixed effects. Standard errors are clustered by stock and date. $\hat{p}^{**} < 0.01$, $\hat{p}^* < 0.05$, $\hat{p} < 0.10$.

A.6 Market robustness

This subsection reports market-side checks supporting the conditional price-relevance interpretation. The tables and figures separate unconditional market lead-lag patterns from state-conditional return association and then vary the definition of the **bN** state.

Table A13: Market outcome local projections: daily **bN** and future market variables

	$h = 1$	$h = 2$	$h = 3$
<i>Panel A. Dependent variable: future return</i>			
Daily bN	-0.0111 (0.0073)	-0.0074 (0.0048)	-0.0017 (0.0041)
Observations	37,758	37,659	37,560
<i>Panel B. Dependent variable: future absolute return</i>			
Daily bN	0.0072 (0.0068)	0.0003 (0.0048)	-0.0010 (0.0041)
Observations	37,758	37,659	37,560
<i>Panel C. Dependent variable: future $\log(1+RV)$</i>			
Daily bN	0.0020 (0.0013)	0.0006 (0.0006)	0.0002 (0.0005)
Observations	37,758	37,659	37,560
Stock fixed effects	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes
Clustering	Stock + date	Stock + date	Stock + date

Notes: Each cell reports the coefficient on daily **bN** from a separate local projection. Controls follow Equation (9); stock and date fixed effects are included. Standard errors are clustered by stock and date. $\hat{p}^{**} < 0.01$, $\hat{p}^* < 0.05$, $\hat{p} < 0.10$.

Table A14: Quartile-tail response of cumulative future returns to current-day negative share

Cumulative horizon	Low-bN tail: Q1 (β_L)		High-bN tail: Q4 (β_H)	
	Coefficient	S.E.	Coefficient	S.E.
$H = 1$	-0.0230**	0.0105	0.0003	0.0084
$H = 2$	-0.0361***	0.0134	0.0131	0.0115
$H = 3$	-0.0418**	0.0171	0.0180	0.0139
$H = 4$	-0.0544***	0.0201	0.0140	0.0170
$H = 5$	-0.0684***	0.0206	0.0102	0.0187
$H = 6$	-0.0651***	0.0224	0.0114	0.0218
$H = 7$	-0.0591**	0.0252	0.0002	0.0237
$H = 8$	-0.0649**	0.0281	-0.0043	0.0264
$H = 9$	-0.0610*	0.0312	-0.0025	0.0291
$H = 10$	-0.0570*	0.0311	-0.0093	0.0306
$H = 11$	-0.0588*	0.0314	-0.0031	0.0324
$H = 12$	-0.0483	0.0306	-0.0069	0.0331
$H = 13$	-0.0504	0.0322	-0.0068	0.0321
$H = 14$	-0.0459	0.0354	-0.0047	0.0328
$H = 15$	-0.0419	0.0355	-0.0060	0.0333
$H = 16$	-0.0432	0.0366	-0.0067	0.0347
$H = 17$	-0.0393	0.0378	-0.0093	0.0363
$H = 18$	-0.0447	0.0403	-0.0122	0.0393
$H = 19$	-0.0463	0.0422	-0.0082	0.0420
$H = 20$	-0.0479	0.0445	-0.0179	0.0435

Notes: Each row reports the slope on current `neg_share` from Equation (11). Low/high states are within-stock bottom/top bN quartiles. Controls include attention, market controls, and ten own-return lags; standard errors are clustered by stock and date. $\hat{p}^{**} < 0.01$, $\hat{p}^* < 0.05$, $\hat{p} < 0.10$.

Table A15: Market-state robustness across alternative bN state definitions

Specification	Horizon	β_L	S.E.	β_H	S.E.	$\beta_H - \beta_L$	S.E.
Quartile-tail Q1 vs Q4	$H = 1$	-0.0230	0.0105	0.0003	0.0084	0.0233	0.0129
Quartile-tail Q1 vs Q4	$H = 5$	-0.0684	0.0206	0.0102	0.0187	0.0786	0.0314
Smooth transition ($\gamma = 1.5$)	$H = 1$	-0.0266	0.0106	0.0072	0.0098	0.0337	0.0156
Smooth transition ($\gamma = 1.5$)	$H = 5$	-0.0601	0.0220	0.0063	0.0242	0.0664	0.0391
Smooth transition ($\gamma = 3.0$)	$H = 1$	-0.0241	0.0097	0.0067	0.0091	0.0308	0.0141
Smooth transition ($\gamma = 3.0$)	$H = 5$	-0.0537	0.0203	0.0023	0.0220	0.0559	0.0349
Median split	$H = 1$	-0.0165	0.0094	0.0056	0.0100	0.0222	0.0143
Median split	$H = 5$	-0.0388	0.0155	0.0019	0.0219	0.0407	0.0296

Notes: Each row reports low-state and high-state slopes, and their difference, from a separate cumulative-return local projection. Smooth specifications use $F(z) = 1/(1 + \exp(-\gamma z))$ with within-stock standardized bN. All specifications include the controls in Equation (11), stock and date fixed effects, and two-way clustered standard errors.

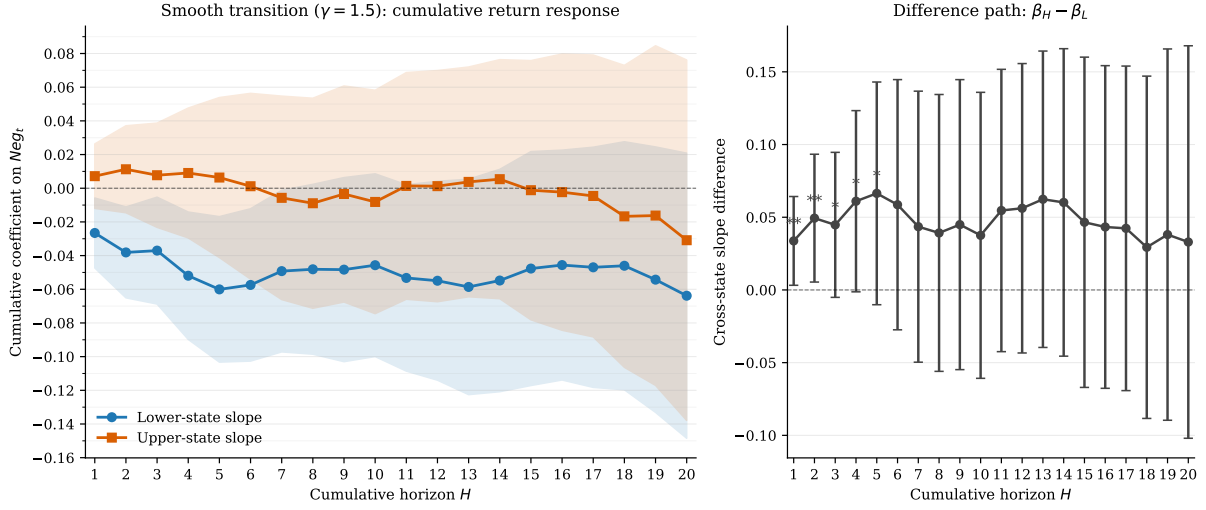


Figure A1: Market-state robustness: smooth-transition specification with $\gamma = 1.5$

Notes: Left: cumulative return response in lower- and upper-bN smooth states. Right: difference path $\beta_H - \beta_L$ across horizons.

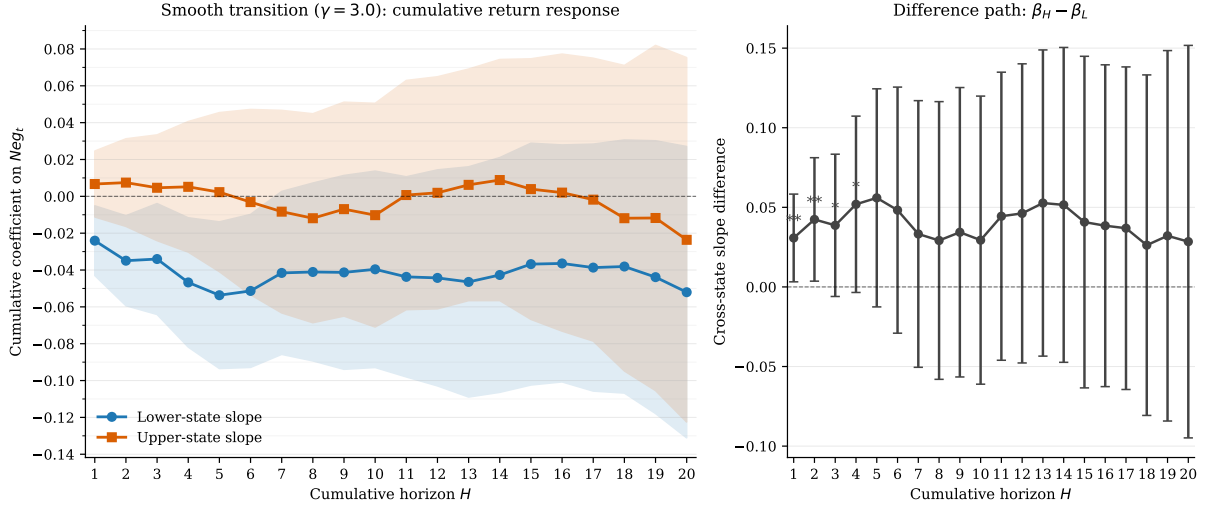


Figure A2: Market-state robustness: smooth-transition specification with $\gamma = 3.0$

Notes: Left: cumulative return response in lower- and upper-bN smooth states. Right: difference path $\beta_H - \beta_L$ across horizons.

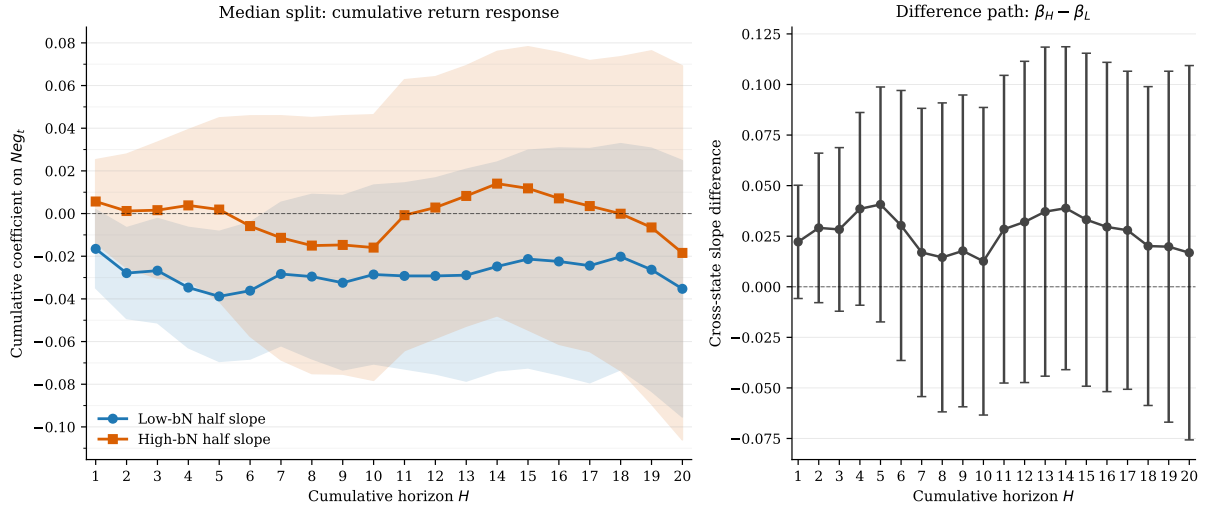


Figure A3: Market-state robustness: within-stock median split

Notes: Left: cumulative return response under the within-stock median split. Right: difference path $\beta_H - \beta_L$ across horizons.